



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

Z-TRANSFORM

- Understand the Z-transform
- Know the ways to transform Z domain.
- Understand the Z-transform properties.

Z - TRANSFORM

1. z-Transform

2. Properties of the Region of Convergence for the z-transform

3. The inverse z-Transform

4. z-Transform Properties

- z-Transform: two-sided, bilateral z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \mathcal{Z}\{x[n]\}$$

$$x[n] \xleftrightarrow{\mathcal{Z}} X(z)$$

- one-sided, unilateral z-transform

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

- If $z = e^{j\omega}$, z-transform is Fourier transform.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- Relationship between z-transform and Fourier transform
 - Express the complex variable z in polar form as

$$z = re^{j\omega}$$

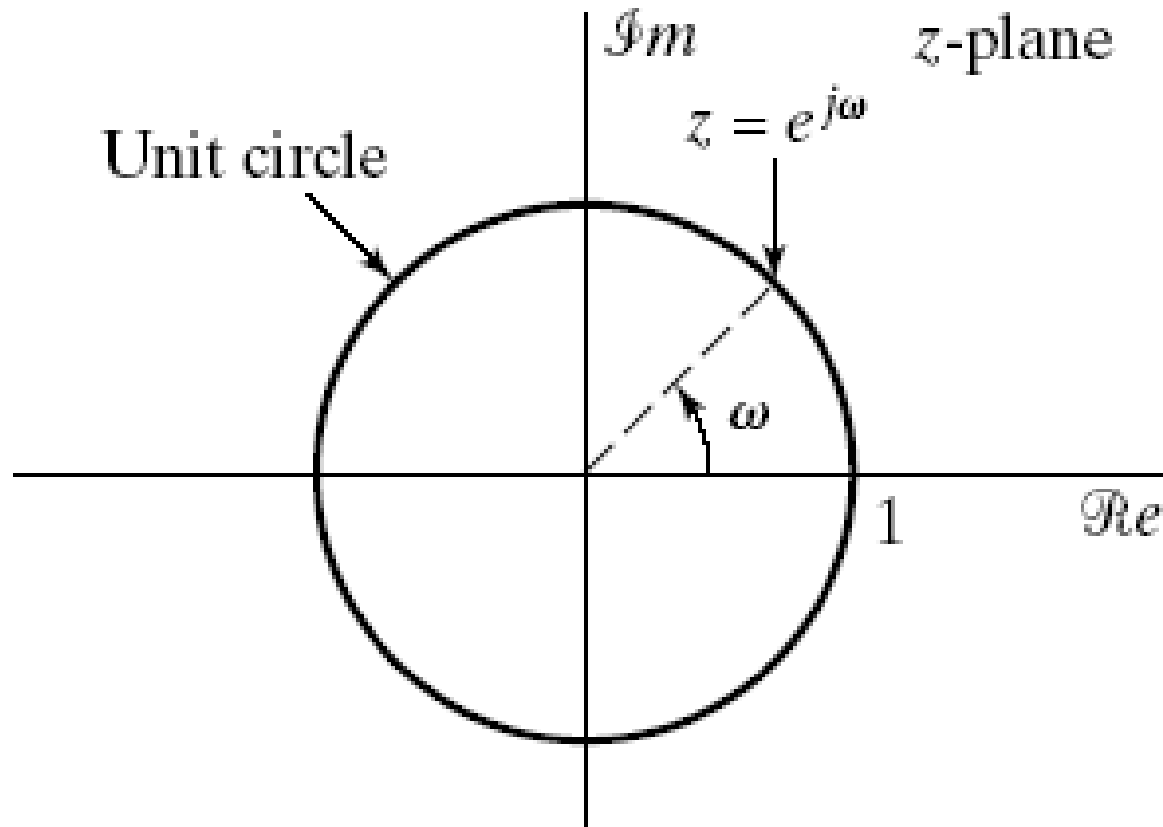
- The Fourier transform of the product of $x[n]$ and the exponential sequence r^{-n}

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n}$$

$$\text{If } r = 1, \quad X(Z) \longrightarrow X(e^{j\omega})$$

- Complex z plane

$$-\pi \leq \omega < \pi \quad \Leftrightarrow \quad \text{unit circle}$$



- Condition for convergence of the z-transform

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} \quad |X(z)| = \sum_{n=0}^{\infty} |x[n]| |z|^{-n} < \infty$$

$$z = re^{j\omega} \longrightarrow X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n}$$

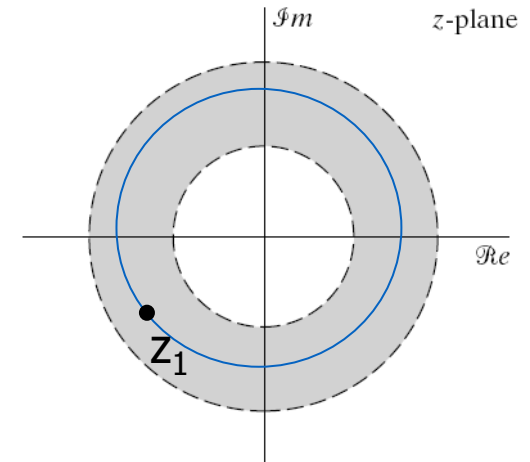
$$|X(re^{j\omega})| \leq \sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty$$

- Convergence of the z-transform for a given sequence depends only on $r = |z|$

- Region of convergence (ROC)
 - For any given sequence, the set of values of z for which the z-transform converges is called the **Region Of Convergence (ROC)**.

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

- if some value of z , say, $z = z_1$, is in the ROC, then all values of z on the circle defined by $|z| = |z_1|$ will also be in the ROC.



- Zero and pole
 - The z-transform is most useful when the infinite sum can be expressed in closed form, usually a ratio of polynomials in z (or z^{-1}).

$$X(z) = \frac{P(z)}{Q(z)}$$

- Zero: The value of z for which $X(z) = 0$
- Pole: The value of z for which $X(z) = \infty$

- Example: Right-sided exponential sequence

Determine the z-transform, including the ROC in z-plane and a sketch of the pole-zero-plot, for the sequence:

Solution:

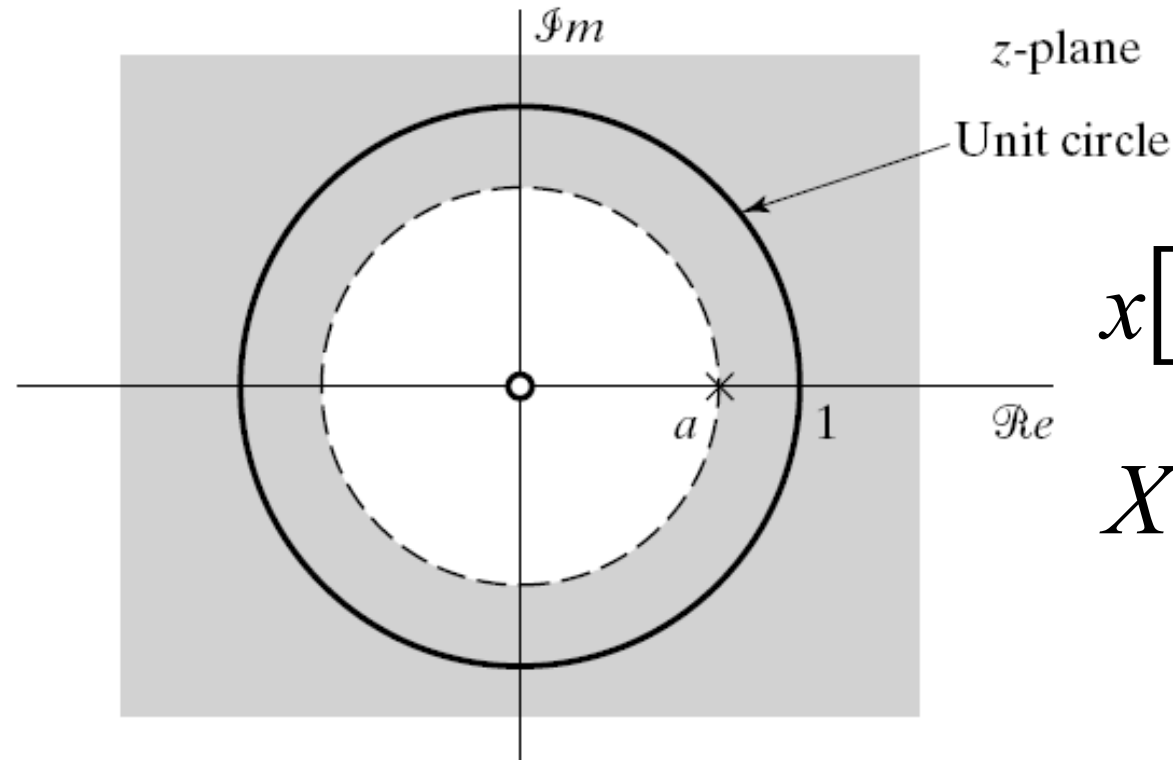
$$x[n] = a^n u[n]$$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

ROC: $|az^{-1}| < 1$ or $|z| > |a|$

zero : $z = 0$

pole : $z = a$



$$x[n] = a^n u[n]$$

$$X(z) = \frac{z}{z - a}$$

○ : zeros

× : poles

Gray region: ROC

- Example: Left-sided exponential sequence

Determine the z-transform, including the ROC, pole-zero-plot, for sequence:

Solution: $x[n] = -a^n u[-n-1]$

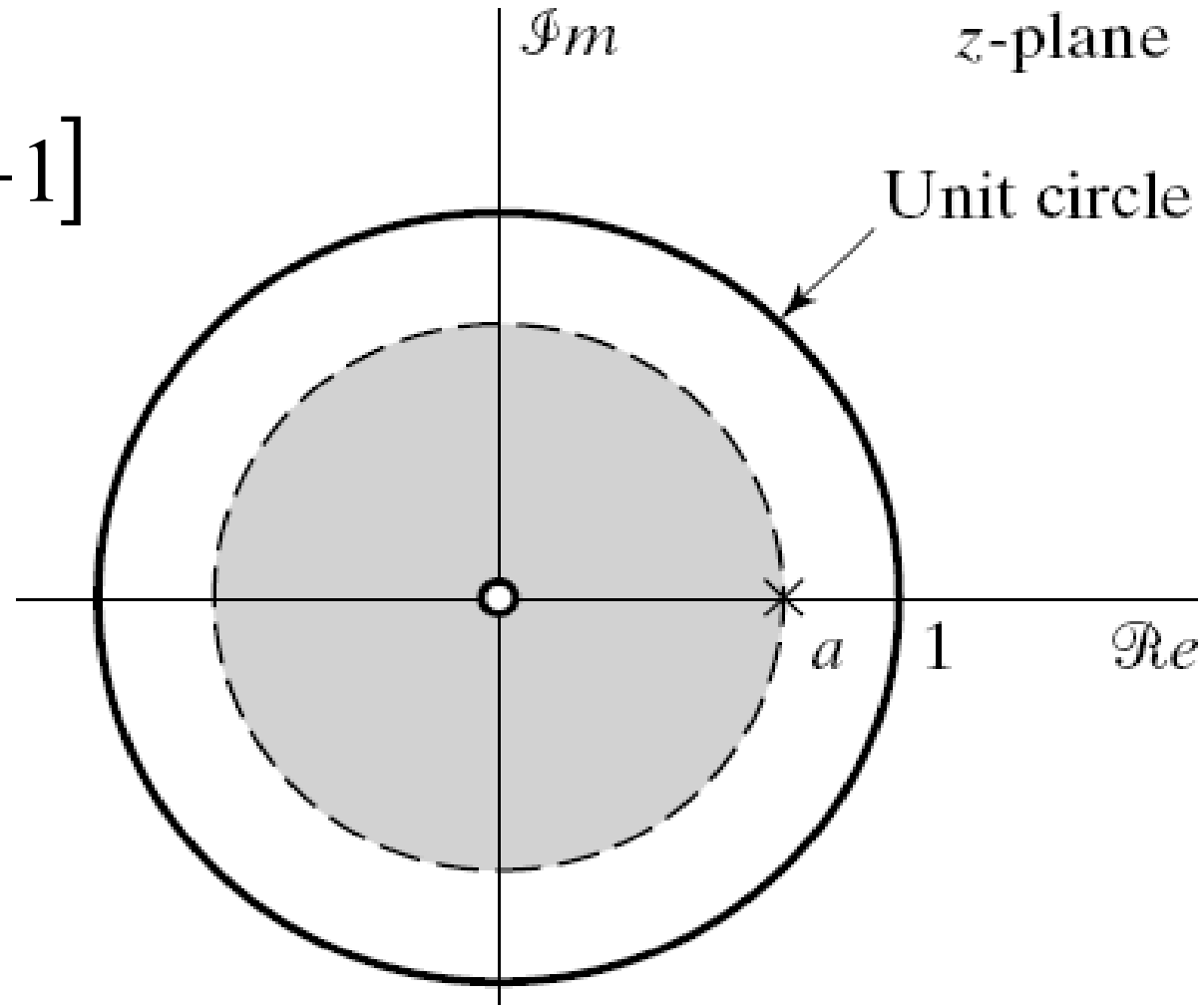
$$\begin{aligned} X(z) &= - \sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n} = - \sum_{n=-\infty}^{-1} a^n z^{-n} \\ &= - \sum_{n=1}^{\infty} a^{-n} z^n = - \sum_{n=1}^{\infty} (a^{-1} z)^n = - \frac{a^{-1} z}{1 - a^{-1} z} = \frac{z}{z - a} \end{aligned}$$

ROC: $|z| < |a|$, *zero* : $z = 0$ *pole* : $z = a$

$$x[n] = -a^n u[-n-1]$$

$$X(z) = \frac{z}{z-a}$$

$$\text{for } |z| < |a|$$



- Example: Sum of two exponential sequences

Determine the z-transform, including the ROC, pole-zero-plot, for sequence:

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$$

Solution:

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} \left\{ \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n] \right\} z^{-n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n] z^{-n} + \sum_{n=-\infty}^{\infty} \left(-\frac{1}{3}\right)^n u[n] z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n + \sum_{n=-\infty}^{\infty} \left(-\frac{1}{3} z^{-1}\right)^n \end{aligned}$$

Z-transform

$$\begin{aligned}
 X(z) &= \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1} \right)^n + \sum_{n=-\infty}^{\infty} \left(-\frac{1}{3} z^{-1} \right)^n \\
 &= \frac{1}{1 - \frac{1}{2} z^{-1}} + \frac{1}{1 + \frac{1}{3} z^{-1}} = \frac{2z \left(z - \frac{1}{12} \right)}{\left(1 - \frac{1}{2} z^{-1} \right) \left(1 + \frac{1}{3} z^{-1} \right)}
 \end{aligned}$$

ROC: $|z| > \frac{1}{2} \text{ and } |z| > \frac{1}{3} \Rightarrow ROC : |z| > \frac{1}{2}$

- Finite-length sequence

$$X(z) = \sum_{n=N_1}^{N_2} x[n] z^{-n}$$

Example:

$$x[n] = \delta[n] + \delta[n-5]$$

$$X(z) = 1 + z^{-5} \quad ROC : |z| > 0$$

- Example: Finite-length sequence

Determine the z-transform, including the ROC, pole-zero-plot, for sequence:

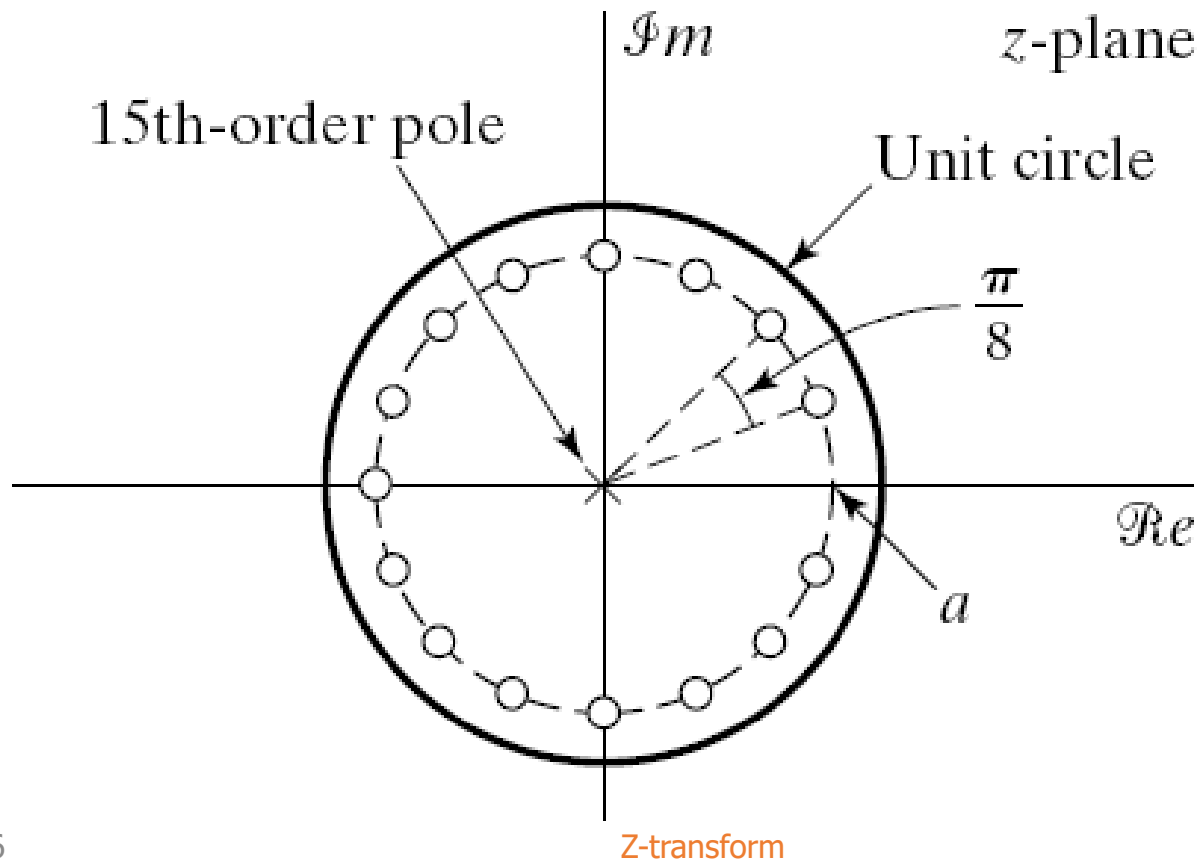
$$x[n] = \begin{cases} a^n, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

Solution:
$$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} (az^{-1})^n$$

$$= \frac{1 - (az^{-1})^N}{1 - az^{-1}} = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z - a} \quad \text{ROC : } |z| > 0$$

$N=16$, a is real

$$X(z) = \frac{1}{z^{N-1}} \frac{z^N - a^N}{z - a} \quad ROC : |z| > 0$$



•z-Transform pairs

$$\delta[n] \leftrightarrow 1, \quad ROC : all \ z$$

$$u[n] \leftrightarrow \frac{1}{1 - z^{-1}}, \quad ROC : |z| > 1$$

$$-u[-n-1] \leftrightarrow \frac{1}{1 - z^{-1}}, \quad ROC : |z| < 1$$

$$\delta[n-m] \leftrightarrow Z^{-m},$$

$$ROC : all \ z \ except \ 0 \ (if \ m > 0) \ or \ \infty \ (if \ m < 0)$$

$$a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad ROC : |z| > |a|$$

$$-a^n u[-n-1] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad ROC : |z| < |a|$$

$$na^n u[n] \leftrightarrow \frac{az^{-1}}{(1 - az^{-1})^2}, \quad ROC : |z| > |a|$$

$$-na^n u[-n-1] \leftrightarrow \frac{az^{-1}}{(1 - az^{-1})^2}, \quad ROC : |z| < |a|$$

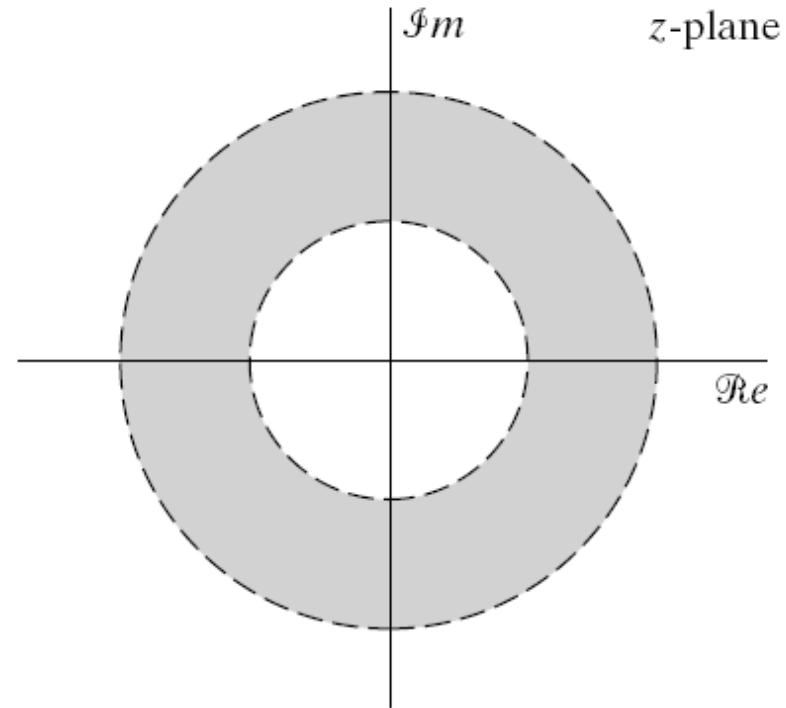
Z - TRANSFORM

1. z-Transform
2. Properties of the Region of Convergence for the z-transform
3. The inverse z-Transform
4. z-Transform Properties

- **Property 1:** The ROC is a ring or disk in the z-plane centered at the origin.

$$0 \leq r_R < |z| < r_L \leq \infty$$

- For a given $x[n]$, ROC is dependent only on $|z|$.



- Property 2: The Fourier transform of converges absolutely if the ROC of the z-transform of $x[n]$ includes the unit circle.
- The z-transform reduces to the Fourier transform when

$$|z| = 1 \quad z = e^{j\omega}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \longrightarrow \quad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- **Property 3:** The ROC cannot contain any poles.
 - $X(z)$ is infinite at a pole and therefore does not converge.

- **Property 4:** If $x[n]$ is a **finite-duration sequence**, i.e., a sequence that is zero except in a finite interval :

$$N_1 \leq n \leq N_2$$

then the ROC is the entire z-plane, except possible $z = 0$ or $z = \infty$

$$X(z) = \sum_{n=N_1}^{N_2} a^n z^{-n}$$

- LTI system Stability, Causality, and ROC
 - Consider a **LTI system** with **impulse response** $h[n]$.
The z-transform of $h[n]$ is called the **system function** $H(z)$ of the LTI system.
 - **Stable system** ($h[n]$ is absolutely summable and therefore has a Fourier transform): **ROC** include **unit-circle**.
 - **Causal system** ($h[n]=0, \text{for } n<0$): right sided

•EXAMPLE: Stability, Causality, and the ROC

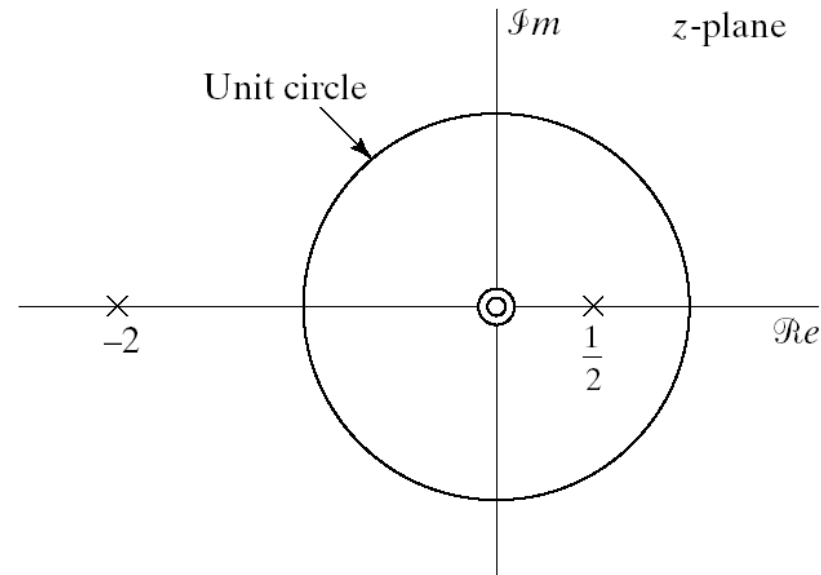
Consider a **LTI system** with **impulse response** $h[n]$. The z-transform of $h[n]$ i.e. the **system function** $H(z)$ has the pole-zero plot shown in Figure. Determine the **ROC**, if the system is:

(1) **stable system**:

(ROC includes unit-circle)

(2) **causal system**:

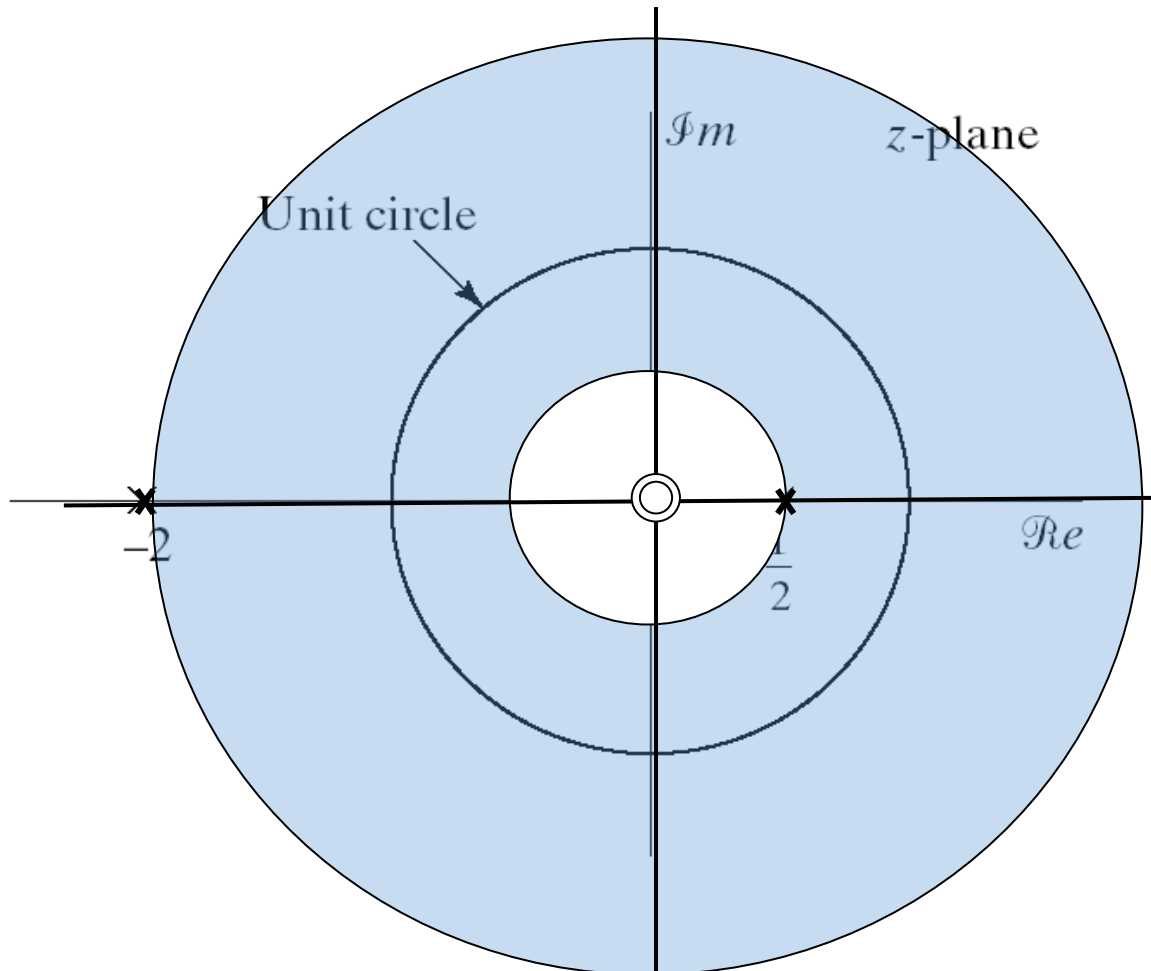
(right sided sequence)



- Solution: (1) stable system (ROC include unit-circle)

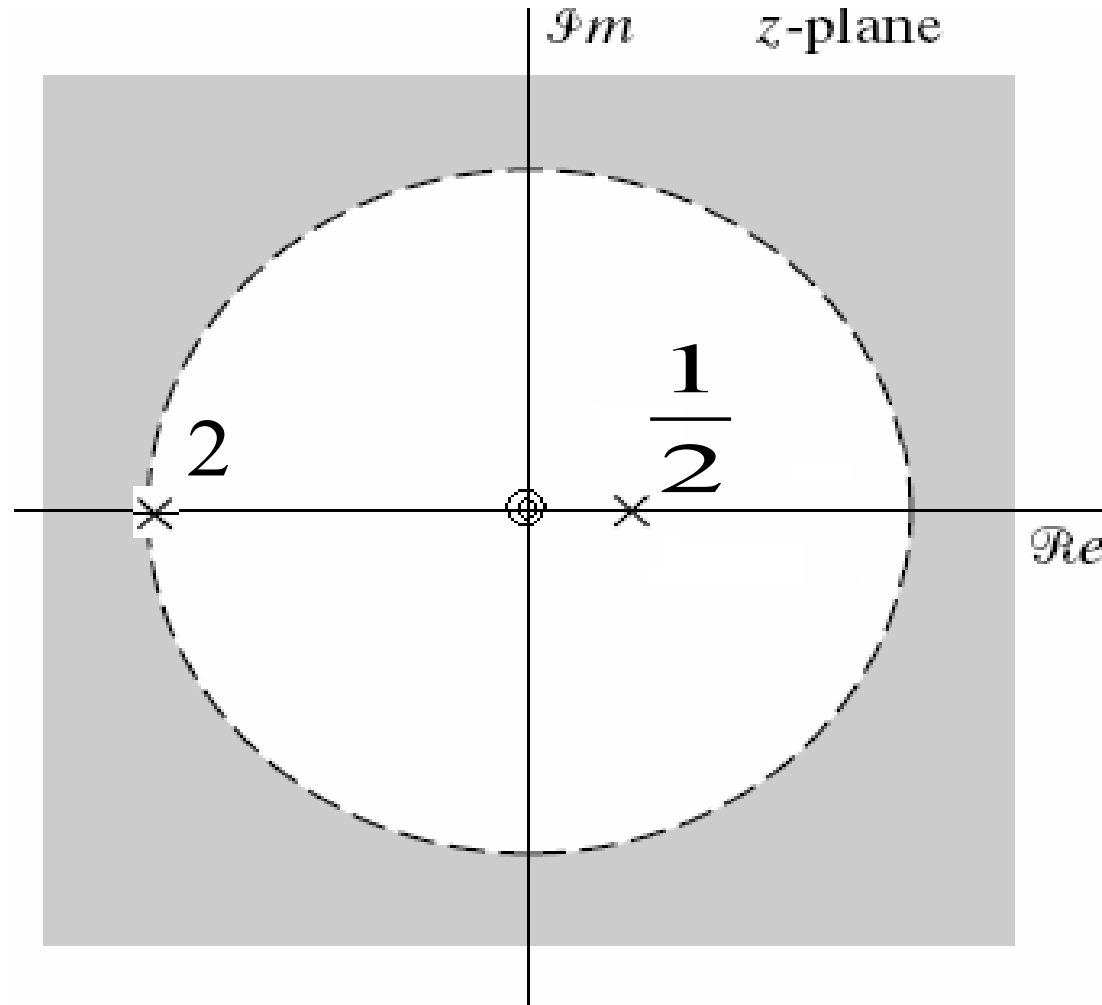
ROC: $\frac{1}{2} < |z| < 2$,

The impulse response is two-sided, and the system is non-causal. **stable**.

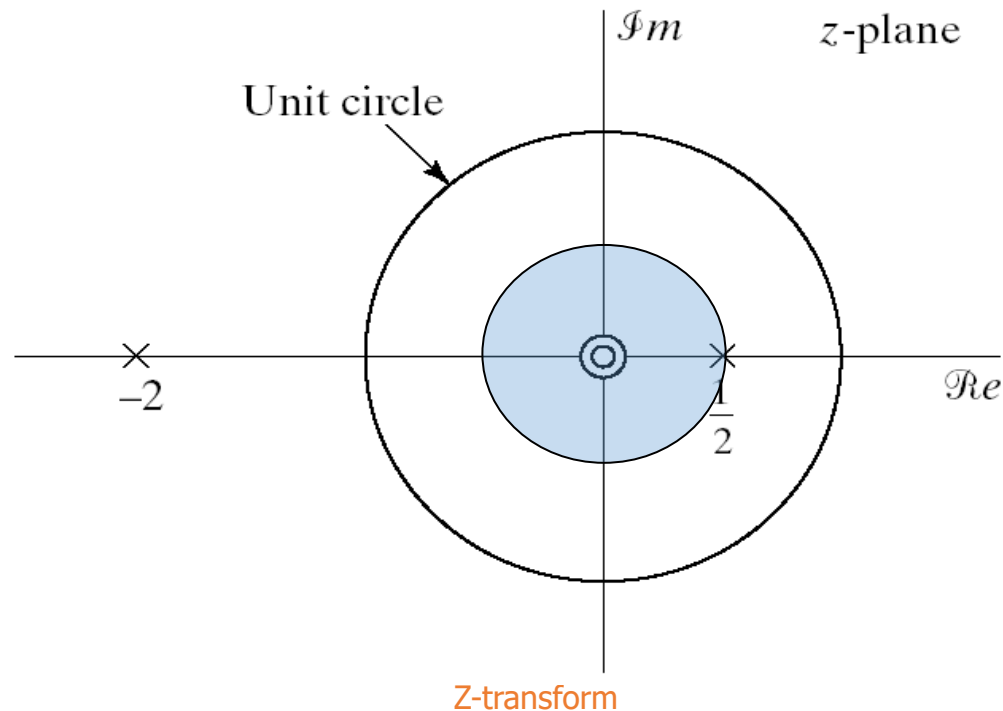


•(2) causal system: (right sided sequence)

- ROC: $|z| > 2$,the impulse response is **right-sided**. system is **causal** but **unstable**.
- A system is **causal** and **stable** if all the poles are **inside the unit circle**.



- ROC: $|z| < \frac{1}{2}$, the impulse response is **left-sided**, system is **non-causal**, **unstable** since the ROC does not include unit circle.



Z - TRANSFORM

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3. The inverse z-Transform
4. z-Transform Properties

- Formal inverse z-transform is based on a Cauchy integral theorem.

$$x_n = \frac{1}{2\pi j} \oint_c X(z) z^{n-1} dz \quad c \in ROC$$

$$= \sum_i \mathbf{Res}[X(z) z^{n-1}]_{z=z_i}$$

- Less formal ways are sufficient and preferable in finding the **inverse z-transform**:
 - Inspection method
 - Partial fraction expansion
 - Power series expansion

- Inspection Method

$$a^n u[n] \xleftrightarrow{z} \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$-a^n u[-n-1] \xleftrightarrow{z} \frac{1}{1-az^{-1}}, \quad |z| < |a|$$

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}$$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1]$$

• Partial Fraction Expansion

$$\begin{aligned}
 X(z) &= \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{z^N \sum_{k=0}^M b_k z^{M-k}}{z^M \sum_{k=0}^N a_k z^{N-k}} = \frac{b_0}{a_0} \frac{\prod_{k=0}^M (1 - c_k z^{-1})}{\prod_{k=0}^N (1 - d_k z^{-1})} \\
 &= \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}} \quad \text{if } M < N
 \end{aligned}$$

where $A_k = (1 - d_k z^{-1}) X(z) \Big|_{z=d_k}$

- Example: Second-Order z-Transform

$$X(z) = \frac{1}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)}, \quad |z| > \frac{1}{2}$$

$$X(z) = \frac{A_1}{\left(1 - \frac{1}{4}z^{-1}\right)} + \frac{A_2}{\left(1 - \frac{1}{2}z^{-1}\right)}$$

$$A_1 = \left(1 - \frac{1}{4}z^{-1}\right)X(z) \Big|_{z=\frac{1}{4}} = -1 \quad A_2 = \left(1 - \frac{1}{2}z^{-1}\right)X(z) \Big|_{z=\frac{1}{2}} = 2$$

$$X(z) = \frac{-1}{\left(1 - \frac{1}{4}z^{-1}\right)} + \frac{2}{\left(1 - \frac{1}{2}z^{-1}\right)}, \quad |z| > \frac{1}{2}$$

$$x[n] = 2\left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

- Partial Fraction Expansion(continue)

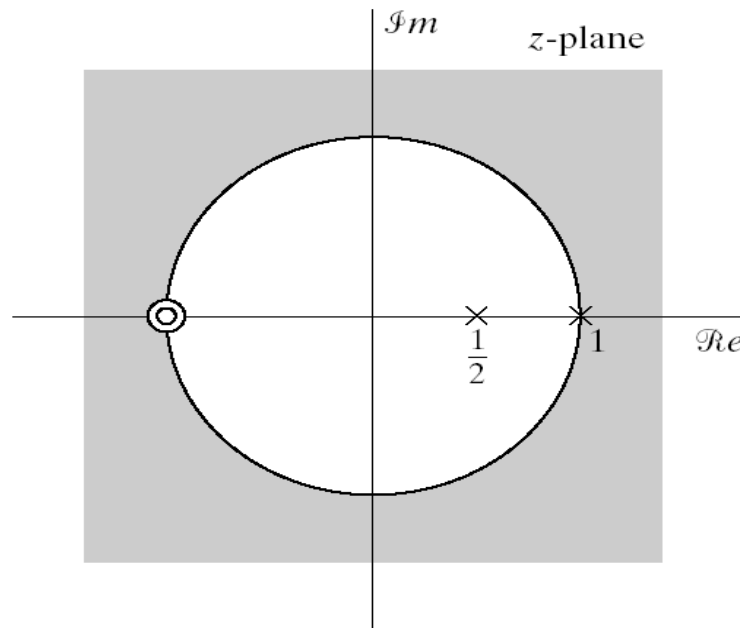
$$X(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \quad \text{if } M \geq N$$

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}}$$

B_r is obtained by long division

•Example: Inverse by Partial Fractions

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{(1 + z^{-1})^2}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})}, \quad |z| > 1$$



$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = B_0 + \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - z^{-1}}$$

$$\frac{\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1}{z^{-2} - 3z^{-1} + 2} = \frac{2}{5z^{-1} - 1}$$

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})}$$

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})} = 2 + \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - z^{-1}}$$

$$A_1 = \left[\frac{-1 + 5z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})} \times \left(1 - \frac{1}{2}z^{-1}\right) \right]_{z=\frac{1}{2}} = -9$$

$$A_2 = \left[\frac{-1 + 5z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})} \times (1 - z^{-1}) \right]_{z=1} = 8$$

$$X(z) = 2 - \frac{9}{\left(1 - \frac{1}{2}z^{-1}\right)} + \frac{8}{(1 - z^{-1})}, \quad |z| > 1$$

$$2 \xleftrightarrow{Z} 2\delta[n] \quad \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)} \xleftrightarrow{Z} \left(\frac{1}{2}\right)^n u[n]$$

$$\frac{1}{(1 - z^{-1})} \xleftrightarrow{Z} u[n] \quad x[n] = 2\delta[n] - 9\left(\frac{1}{2}\right)^n u[n] + 8u[n]$$

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$$x[n] \xleftrightarrow{Z} X(z), \quad ROC = R_x$$

•Linearity

$$x_1[n] \xleftrightarrow{Z} X_1(z), \quad ROC = R_{x_1}$$

$$x_2[n] \xleftrightarrow{Z} X_2(z), \quad ROC = R_{x_2}$$

$$ax_1[n] + bx_2[n] \xleftrightarrow{Z} aX_1(z) + bX_2(z)$$

$$ROC \text{ contains (may more than) } R_{x_1} \cap R_{x_2}$$

•Time Shifting

$$x[n - n_0] \xleftrightarrow{Z} z^{-n_0} X(z)$$

- ◆ n_0 is an integer
- ◆ n_0 is positive, $x[n]$ is shifted right
- ◆ n_0 is negative, $x[n]$ is shifted left

$$ROC = R_x \left(\begin{array}{l} \textit{except for the possible addition} \\ \textit{or deletion of } z = 0 \text{ or } z = \infty \end{array} \right)$$

•Example: Shifted Exponential Sequence

$$X(z) = \frac{1}{z - \frac{1}{4}}, \quad |z| > \frac{1}{4} \xleftrightarrow{z} \left(\frac{1}{4}\right)^n u[n]$$

$$X(z) = \frac{z^{-1}}{1 - \frac{1}{4}z^{-1}} = -4 + \frac{4}{1 - \frac{1}{4}z^{-1}}$$

$$x[n] = -4\delta[n] + 4\left(\frac{1}{4}\right)^n u[n]$$

$$X(z) = z^{-1} \left(\frac{1}{1 - \frac{1}{4}z^{-1}} \right)$$

$$x[n] = \left(\frac{1}{4}\right)^{n-1} u[n-1]$$

- Multiplication by an Exponential sequence

$$z_0^n x[n] \xleftrightarrow{Z} X(z / z_0), \quad ROC = |z_0| R_x$$

if R_x is $r_R < |z| < r_L$, $|z_0| R_x$ is $|z_0| r_R < |z| < |z_0| r_L$

$$\text{if } z_0 = e^{j\omega_0} \quad z = e^{j\omega}$$

$$e^{j\omega_0 n} x[n] \xleftrightarrow{F} X(e^{j(\omega - \omega_0)}): AM$$

- Differentiation of $X(z)$

$$nx[n] \xleftrightarrow{z} -z \frac{dX(z)}{dz}, \quad \textcolor{red}{ROC = R_x}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$\begin{aligned} -z \frac{dX(z)}{dz} &= -z \sum_{n=-\infty}^{\infty} (-n) x[n] z^{-n-1} \\ &= \sum_{n=-\infty}^{\infty} nx[n] z^{-n} = Z \{ nx[n] \} \end{aligned}$$

- Convolution of Sequences

$$x_1[n] * x_2[n] \xleftrightarrow{Z} X_1(z) X_2(z), \quad \text{ROC contains } R_{x_1} \cap R_{x_2}$$

$$y[n] = x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k]$$

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} y[n] z^{-n} = \sum_{n=-\infty}^{\infty} \left\{ \sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k] \right\} z^{-n} \\ &= X_1(z) X_2(z) \end{aligned}$$

- Definition z-transform
- Properties of the Region of Convergence for the z-transform
- Many ways to perform the inverse z-Transform
- Important Properties of Z-transform



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THANK YOU